

Homework 1

Due Sep 26, start of class

 Write down your solutions and submit them as a single PDF to loss0f0706@gmail.com.
Your solutions must be handwritten.**Problem 1. From words to a model****(40 points)**

A transport aircraft is being loaded for a resupply flight. Two cargo types, with value, weight and volume given per pallet:

	ammunition	fuel
value (pts)	4	5
weight (t)	3	2
volume (m ³)	1	3

The aircraft carries at most 60 t and 48 m³, and orders require that at least 3 pallets of ammunition fly. Cargo is divisible — fractional pallets are allowed. The goal is to maximize total value.

- (a) **(8 pts)** Define the decision variables and write the problem as a scalar LP — objective and constraints, one line each, with a few words saying what each constraint enforces. Write each line the way the story gives it — no rearranging yet.

hint: this is the toy LP from lecture with different numbers, plus one extra row for the minimum ammunition order — write it in the same shape.

- (b) **(8 pts)** Now put that LP in the Week 2 canonical vector form $\max c^\top x$ s.t. $Ax \leq b$: write out c , A , b with numbers in them, and report the dimensions of A .

hint: $x \geq 0$ is equivalent to $-x \leq 0$, and $x \geq 3$ to $-x \leq -3$.

- (c) **(14 pts)** Solve it by picture — with two decision variables, the problem can be drawn in the 2D plane! Draw the feasible region in Desmos (desmos.com/calculator). **Include a screenshot of your Desmos plot.** Then list every vertex with its coordinates and its objective value, and report the optimal solution and its optimal value.

hint: an LP optimum always sits at a vertex, so checking the corners is enough — no need to test anything inside.

- (d) **(10 pts)** Now let the computer check your picture. Implement the LP with `gurobipy` and solve it, starting from the Week 2 Colab notebook.¹ Include a screenshot of your code and its printed output, using exactly this form:

```
print(f"optimal solution ammunition is {x_ammo.X}")
print(f"optimal solution fuel is {x_fuel.X}")
print(f"optimal objective value is {model.ObjVal}")
```

Check that Gurobi agrees with the answer you read off your drawing in (c).

hint: Gurobi minimizes unless told otherwise — pass `GRB.MAXIMIZE` to `setObjective`.

Problem 2. Convex sets**(20 points)**

A set $S \subseteq \mathbb{R}^n$ is convex when

$$\lambda x + (1 - \lambda)y \in S \quad \text{for all } x, y \in S \text{ and all } \lambda \in [0, 1]$$

— the *segment test* from lecture: the straight segment between any two points of S stays inside S .

¹ https://colab.research.google.com/github/hankpark0706/AD7031/blob/main/notebooks/week02_linear_program.ipynb

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- (a) **(10 pts)** Draw two convex sets in $\mathbb{R}^2$ whose *union* is not convex, marking two points and the point on their segment that escapes. Then *try* to draw two convex sets whose *intersection* is not convex. After a few honest attempts, say in one sentence what you believe is true about intersections — pictures are enough here, no proof needed.

hint: for the union, two convex sets that do not touch each other already do it.

- (b) **(10 pts)** Show that a polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ is a convex set. This is the feasible set you drew in Problem 1.

hint: of course, P is n -dimensional, so you cannot draw it on paper any more. Use the definition: take $x, y \in P$ and check that $A(\lambda x + (1 - \lambda)y) \leq b$.

Problem 3. Convex functions

(20 points)

A function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is convex when

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad \text{for all } x, y \in \mathbb{R}^N \text{ and all } \lambda \in [0, 1].$$

- (a) **(10 pts)** Suppose $f_k : \mathbb{R}^N \rightarrow \mathbb{R}$, $k \in [K]$, are convex functions. Show that the following functions are convex:

- $g(x) = \sum_{k \in [K]} f_k(x)$
- $g(x) = \max_{k \in [K]} \{f_k(x)\}$

hint: use the definition above — each proof is about one line. Citing Week 2's convexity-preserving operations instead is also fine.

- (b) **(5 pts)** Show that $f(x) = x^2$ is a convex function.

hint: use the definition above — again, it is a single-line proof.

- (c) **(5 pts)** Show that $f(x) = \|x\|_2^2$ is a convex function.

hint: $\|x\|_2^2$ is just $x_1^2 + x_2^2 + \dots + x_N^2$ — see (a) again.

Problem 4. Markowitz portfolio

(20 points)

Three funds — bond, tech, energy. Next year's returns are not known; all we have is their mean vector μ and covariance matrix Σ :

$$\mu = \begin{pmatrix} 0.06 \\ 0.10 \\ 0.15 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.010 & 0.001 & 0.000 \\ 0.001 & 0.040 & 0.015 \\ 0.000 & 0.015 & 0.090 \end{pmatrix}$$

The Markowitz model splits the budget across the three funds so as to minimize risk, subject to a required mean return r :

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & x^\top \Sigma x \\ \text{s.t.} \quad & \mu^\top x \geq r \\ & \mathbf{1}^\top x = 1 \\ & x \geq 0 \end{aligned}$$

- (a) **(15 pts)** Choose your own minimum mean return r that you want — for example 0.01, or 0.20. Implement the model with `gurobipy`, starting from this week's Colab notebook,² and print your result in exactly this form:

```
print(f"optimal solution bond is {x_bond.X}")
print(f"optimal solution tech is {x_tech.X}")
print(f"optimal solution energy is {x_energy.X}")
print(f"optimal objective value is {model.ObjVal}")
```

² https://colab.research.google.com/github/hankpark0706/AD7031/blob/main/notebooks/week03_markowitz.ipynb

Is your problem feasible for the r you chose? Answer whichever case you land in:

- **Infeasible.** Explain why the problem cannot be solved — in plain Korean, with no mathematics.
- **Feasible.** Raise r little by little until the problem stops being feasible. Report the r at which that happens, and explain why it happens — again in plain Korean, with no mathematics.

hint: you cannot be too greedy.

- (b) **(5 pts)** Is the Markowitz problem a convex problem? If it is, explain why.

hint: a convex problem is one whose objective f is convex and whose feasible set is a convex set — Problems 2 and 3 did most of that work already.